

Experiment 4: Insertion Sort – Playlist Reordering by Recently Added

Aim

To implement the **Insertion Sort** algorithm in Python to insert a newly added song into its correct position in a playlist sorted by duration.

Problem Statement

A music application stores songs in ascending order of duration. When a new song is added, it should be inserted into the correct position without sorting the entire playlist again.

Algorithm

Step 1: Start.

Step 2: Read the number of songs.

Step 3: Read the song names and durations.

Step 4: Apply Insertion Sort based on song duration.

Step 5: Shift songs with longer durations to the right.

Step 6: Insert the current song in its correct position.

Step 7: Display the sorted playlist.

Step 8: Stop.

Source Code (Python)

```
main.py
1  # Playlist Reordering using Insertion Sort
2
3  n = int(input("Enter number of songs: "))
4  playlist = []
5
6  for i in range(n):
7      name = input("Enter song name: ")
8      duration = int(input("Enter duration (seconds): "))
9      playlist.append([name, duration])
10
11  for i in range(1, n):
12      key = playlist[i]
13      j = i - 1
14
15      while j >= 0 and playlist[j][1] > key[1]:
16          playlist[j + 1] = playlist[j]
17          j -= 1
18
19      playlist[j + 1] = key
20
21  print("\nSorted Playlist:")
22  for song in playlist:
23      print(song[0], "-", song[1], "seconds")
```

Output

Output

```
Enter number of songs: 4
Enter song name: intro
Enter duration (seconds): 120
Enter song name: chill beat
Enter duration (seconds): 210
Enter song name: long jam
Enter duration (seconds): 340
Enter song name: quick track
Enter duration (seconds): 180

Sorted Playlist:
intro - 120 seconds
quick track - 180 seconds
chill beat - 210 seconds
long jam - 340 seconds

=== Code Execution Successful ===
```

Complexity Analysis

Aspect	Complexity
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Time Complexity	$O(n^2)$
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Space Complexity	$O(1)$
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Result

The Insertion Sort algorithm was successfully implemented to reorder the playlist by song duration. The program correctly inserted each song into its proper position and displayed the playlist in ascending order of duration.